

.gitignore patterns

Simulators and builds leave junk behind, logs, waveform dumps, object files,
whole output folders

Ignore means “do not track”

- A `.gitignore` file lives at the repo root and lists patterns
- Matching untracked paths disappear from status and will not be added by mistake
- Typical design patterns ignore a build directory, log files, and waveform dumps
- Remember: ignore rules apply to untracked files

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

.gitignore pattern lab

Write ignore rules for sim/build junk (build/, *.log, *.vcd) and see which paths stay trackable.

Starter example: Ignore build/, *.log, *.vcd, and object/editor junk — source under rtl/ and tb/ stays tracked.

Load starter example

Challenges 0 / 22 cleared

Quiz: *.log: *.log ignores files by? Answer: extension

Answer

Show hintCheckNextIdle

Quiz: *.logQuiz: build/Ignore buildKeep rtlApply starterCount vcdOnly *.vcdbuild/ onlyEmpty rulesNegationQuiz: !Root-onlyQuiz: leading /Object filesMakefile okWrite logs/Filter ignoredQuiz: purposeIgnore swpIgnore out/Keep scriptsQuiz: already tracked

.gitignore

Starter HDLOnly *.vcdbuild/ onlyEmptyNegation demoRoot-only wave

```
# Chip project ignore rules
build/
out/
obj/

# Logs and waveforms
```

Path status

11 ignored · 9 tracked-ok · 20 paths

☒ all☐ ignored☐ tracked-ok

path	status	rule
Makefile	tracked-ok	-
README.md	tracked-ok	-
.gitignore	tracked-ok	-
rtl/top.v	tracked-ok	-

.gitignore lab starter

learn_git — gitignore lab

Real Git practice

```
wsl - module03-gitignore-lab/examples/gitignore

# Track A - real Unix
# real shell session (Track A)
# practice repo: /tmp/git-gi-3rwHuF

$ git init
Initialized empty Git repository in /tmp/git-gi-3rwHuF/.git/

$ mkdir -p build

$ touch build/output.tmp

$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file> ..." to include in what will be committed)
    build/

nothing added to commit but untracked files present (use "git add" to track)

$ printf 'build/\n*.log\n*.vcd\n' > .gitignore

$ cat .gitignore
build/
*.log
*.vcd

$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file> ..." to include in what will be committed)
    .gitignore

nothing added to commit but untracked files present (use "git add" to track)

$ ls build/output.tmp
build/output.tmp
```

Real shell — status before and after .gitignore

Real Git practice — try these

```
# git init — create a new repository here  
git init
```

```
# mkdir -p build — make a build output folder  
mkdir -p build
```

```
# touch build/output.tmp — fake a generated file  
touch build/output.tmp
```

```
# git status — see the untracked build output  
git status
```

```
# printf ... > .gitignore — ignore the build/ directory  
printf 'build/\n*.log\n*.vcd\n' > .gitignore
```

```
# git status — build output should no longer appear
```

Pitfalls to watch

- Do not commit large regenerated files and then add ignore later
- Do not ignore your RTL or testbench sources by accident with a broad star pattern
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load the starter and finish a few challenges
- On real Git, copy or adapt the sample ignore file from this module's examples into your
- When you are ready